PADELLENS

Pro Tour Insights + My Match Log

A data visualization tool for amateur padel players.

Sensei · Data Visualization for Sport · Sports Engineering MSc
Politecnico di Milano · AY 2025/2026

OVERVIEW

How this talk is organized

1	**The Brief**	Sport domain, audience, editorial angle
2	**Working with Data**	Sources, schema, exploration
3	**UX Design**	Wireframes, principles, heuristic eval
4	**Data Representation**	Chart choices, color, interactivity
5	**Technical Implementation**	Python, Streamlit, architecture
6	**Demo & Conclusions**	Live walkthrough, future work

PART 1

The Brief

Context & Vision · Stage 1 of Kirk's data visualization process

1.1 SPORT DOMAIN

Why padel?

Padel is exploding.

1.1M+ federated players in Italy (2024)

Up from ~30,000 in 2019 (FIP data).

Pros have analytics.
Amateurs have a group chat.

Premier Padel rankings, ball-tracking, broadcast overlays - none of that reaches the club player.

THE GAP

What amateurs feel

"I always lose in the third set."

"I play better with Luca."

"My bandeja is OK, my smash isn't."

What they can confirm

Nothing.

That gap is the project.

Who's at the other end of the screen?

PROBLEM STATEMENT

Amateur padel players want to see patterns in their own game that they can't feel - but no simple tool turns a player's match log into honest visual feedback.

PRIMARY USER

Marco · the club player

- 22-40 years old
- Plays 1-3 times per week at his local club
- 6 months to 5 years of experience
- Will spend 90 seconds logging on his phone
- Wants to improve · is not a pro · has no coach

SECONDARY USER

Luca · the amateur coach

- Gives weekly 1-hour lessons
- Needs a 2-min summary of each student before training
- Cares about shot balance, error trends, partner effect
- Reads on a laptop, not a phone

Angle · Framing · Focus (Kirk)

ANGLE

Reflective self-analysis over time

Temporal · Categorical · Comparative.

Spatial overlays (court heatmaps) deliberately excluded - we don't have sensor data.

FRAMING

What an amateur can log on a phone

Inside: matches, sets, shot tallies, partners, surface.

Outside: ball-trajectory, biometrics, opponent scouting.

FOCUS

One number above the fold

Last-10-matches win rate.

Plus: net shot balance, plain-language insight ("You win 80% with Luca").

Scored on Kirk's four factors

Timeliness	3/3	Opens after a match or before the next one - actionable now.
Interestingness	3/3	Confirms or contradicts gut feelings - both outcomes are interesting.
Pertinence	3/3	Literally the user's own data - maximum personal connection.
Sufficiency	2/3	Deliberate trade-off: enough to inform, not so much it overwhelms.

“You don't have to feel your game - you can see it.”

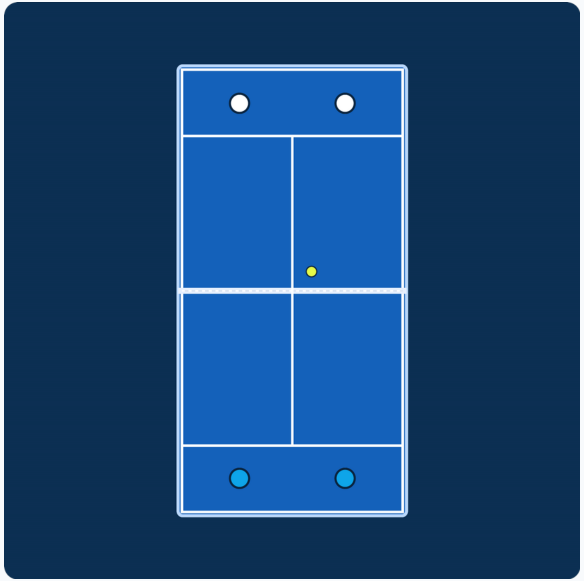
PART 2

Working with Data

Acquisition · Transformation · Exploration

Premier Padel: the elite circuit behind our data

The official global professional padel tour (launched 2022, run by the FIP and Qatar Sports Investments). In 2024 it absorbed the former World Padel Tour, unifying the pro game. My app's Pro Tour layer pulls its rankings and match results from this circuit.



Premier Padel court (top-down). Swap in an official photo if you have rights.

~26 tournaments per season, on 5 continents (Majors + P1 + P2 + Finals)

Majors	2,000 pts	Top tier - 4 a year (Qatar, Italy, France, Mexico)
P1	1,000 pts	Second tier - e.g. Madrid, Milano, Dubai, London
P2	~500 pts	Entry tier - e.g. Brussels, Rotterdam, Germany
Finals	1,500 pts	Season finale - top 16 pairs, Barcelona

Below Premier Padel: the CUPRA FIP Tour feeder circuit (Platinum / Gold / Silver / Bronze), where emerging pairs earn points and climb toward P2 and P1.

2.1 ACQUISITION

Two layers, two sources

LAYER A · PRO TOUR (REAL)

Padel API · padelapi.org

- Premier Padel rankings + match results
- Free tier: 50k req/month, 6 months of data
- Token auth, REST + JSON
- *Paid only: full history · live point-by-point · shot stats*

BACKUP

- Kaggle Padel Tennis WC dataset
- Premier Padel website scrape
- Cached CSV for offline demo

LAYER B · PERSONAL LOG

Seed data + user-generated

No public amateur padel match data exists.
That absence is part of why this project exists.

APPROACH

- Realistic seed log demonstrates the app
- Real users replace it via Log Match form
- Transparently disclosed - not hidden

Three CSVs, tidy schemas

pro_players.csv — 30 rows

```
player_id
name
country
side (D/R)
ranking_points
+ 3 more
```

pro_matches.csv — 276 rows

```
match_id
date · tournament
round
team1_p{1,2}
team2_p{1,2}
winner_team
set{1,2,3}
surface
duration_min
```

my_matches.csv — 40+ rows

```
match_id · date
partner · opponents
club · surface
set scores (per set)
winners_{shot} ·
errors_{shot}
result · duration · notes
```

Transformations (cached in Streamlit): score-string parsing → numeric · derived 'won' & 'net_<shot>' · groupby on partner / surface / shot · 5-match rolling win rate.

What the data told me before I drew anything

PRO TOUR

68%

straight-set rate

Matches the real Premier Padel rate (~68-72%). Means: third-set analytics are rare on the pro side. Don't waste space on them.

PARTNER

80% vs 20%

Luca vs Davide

60-point spread across the same player. This is the single most surprising actionable fact in the data.

FATIGUE

−40 pp

drop in 3-set matches

Two-set win rate is 80%, three-set is 40%. The fade is real, and it's the biggest learnable pattern.

PART 3

UX Design

Wireframes · Principles · Heuristic evaluation

From paper sketch to working app

1. **Paper sketches**

   30 min. Just box drawings - which pages exist.

2. **Low-fi wireframes**

   Figma · 5 SVGs · grayscale layout only.

3. **Mid-fi pass**

   Same layout, real chart shapes drawn in.

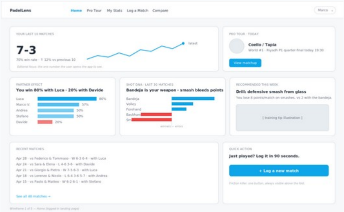
4. **Streamlit build**

   High-fi == working app. No click-prototype.

Why I stopped at low-fi: *The deliverable is a Streamlit app, and going from a Streamlit-shaped wireframe to a Streamlit-built app is a one-step jump. Click-prototypes are theatre.*

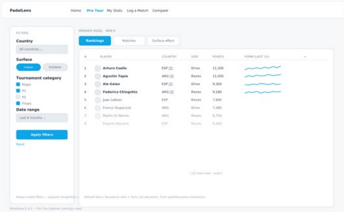
3.2 WIREFRAMES

Five pages, each built around one user task



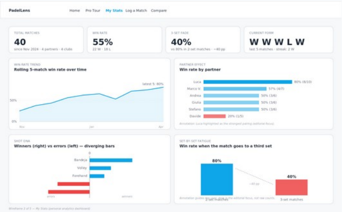
Home

Last 10 + partner + shot DNA



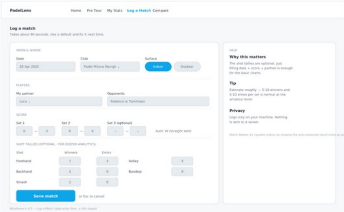
Pro Tour

Rankings + filters



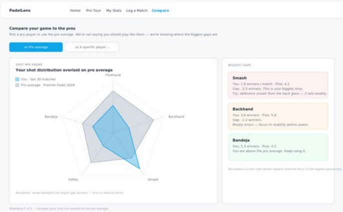
My Stats

Personal dashboard



Log Match

90-second data entry



Compare

Radar vs pro reference

Each page answers ONE task from the brief.

Home → "How am I doing right now?" · **Pro Tour →** "Explore the pro game" · **My Stats →** "Why am I winning/losing?" · **Log Match →** "Add today's match" · **Compare →** "How big is the gap to a pro?"

Six rules applied throughout

1 **One number above the fold**

The app is opened in bursts. 42-pt KPI on Home.

2 **Recognition over recall**

Filters always visible - never behind a button.

3 **Default to action**

+ Log a match is the most prominent CTA on Home.

4 **Reduce data-entry friction**

Shot tallies optional. Score + partner is enough.

5 **Honest visual hierarchy**

Red only for negative signal. No gratuitous color.

6 **Annotate, don't decorate**

Every chart has at least one in-canvas annotation.

Two real changes from Nielsen's 10

	BEFORE		AFTER
#5 Error prevention	Original form let me enter Set-3 scores even when sets 1-2 were already decided (6-0, 6-0).	→	Set-3 inputs are disabled until sets 1-2 are split 1-1.
#6 Recognition over recall	First sketch had filters hidden behind a 'Filters' button - user has to remember what they filtered by.	→	Filters live in an always-visible sidebar on the Pro Tour page.

PART 4

Data Representation

Chart choices · Color · Interactivity

4.1 CHART INVENTORY

Nine chart types · each answers one EDA finding

CHART	WHERE	ANSWERS
KPI + sparkline	Home	Last-10 win rate
Horizontal bar	Home / My Stats	Partner win rate
Diverging bar	Home / My Stats	Shot DNA (winners − errors)
Ranking table + row sparkline	Pro Tour	Top players + form
Small-multiples	Pro Tour	Surface effect
Rolling-mean line	My Stats	Win-rate trend
Set-by-set columns	My Stats	Fatigue (2-set vs 3-set)
Radar overlay	Compare	You vs pro shot mix
Inline annotations	All pages	Headline numbers

Rejected: pie charts, court heatmaps, Sankeys, 3D bars, animated rankings. Each lost head-to-head against a simpler alternative.

4.2 CASE STUDY

Why a diverging bar for shot DNA?

THE QUESTION

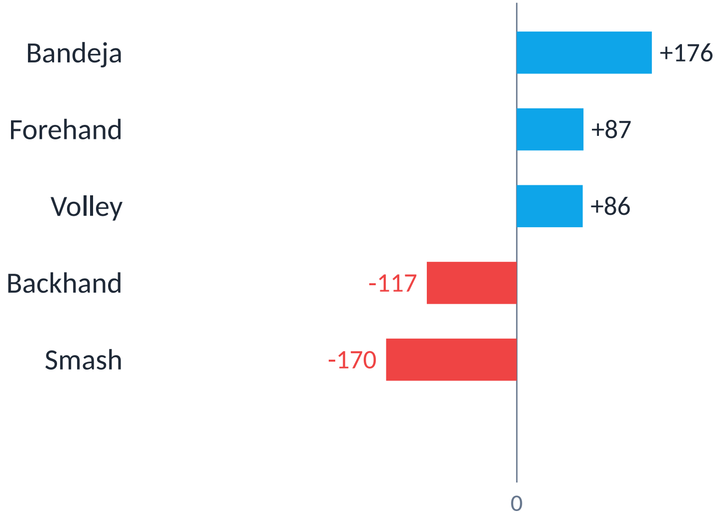
Which shots win me points, which bleed them?

The question is signed. Some shots are net positive, others net negative.

CONSIDERED:

Pie of shot share - hides the zero line.
Stacked bar - needs eye traversal.
Two side-by-side bar charts - same problem.
Diverging bar - chosen.

YOUR SHOT DNA · 40 MATCHES



Zero line dead center · sign readable at a glance · winners and errors share the bar.

4.3 COLOR

Editorial use, not decoration

Blue	#0EA5E9	*Positive / interactive / default*
Red	#EF4444	*Negative signal requiring action*
Gray	#94A3B8	*Pro reference frame*
Ink	#1F2937	*Body text*
Muted	#6B7280	*Secondary text*
BG	#F8FAFC	*Page background*

ACCESSIBILITY CHECKS

WCAG AA body-text contrast 12.6:1 ✓ · Blue/red palette: deuteranopia & protanopia safe ✓ · No green/red pairs · **Every red mark is also negatively signed numerically.**

Kirk's last two design layers

INTERACTIVITY

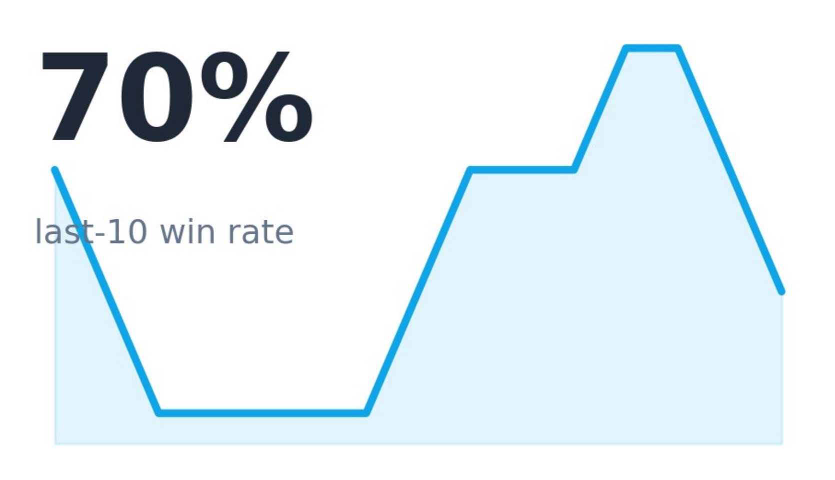
- Sidebar filters (country, surface, category, dates)
- Row-click drill from rankings to player detail
- Time-range toggle on My Stats (30/60/90/all)
- Hover tooltips: precision on demand
- Live auto-result preview while typing a score

ANNOTATION

- Static in-canvas: the headline ("-40 pp")
- Dynamic hover: the detail (exact value, date)
- Captions outside the chart: WHAT
- Callouts inside the chart: WHY you should care
- Color-coded gap cards on Compare (red→amber→green)

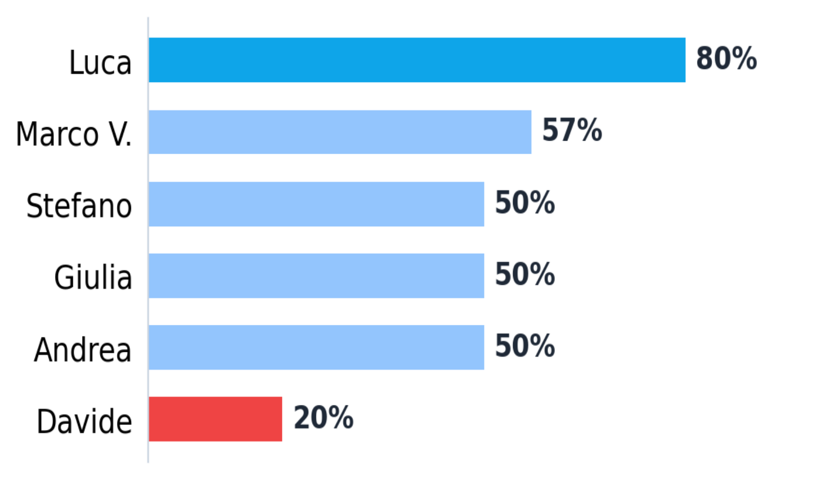
The charts in the app (1/4)

Win-rate KPI + sparkline



The single headline number the user opens the app for, with a sparkline showing recent momentum.

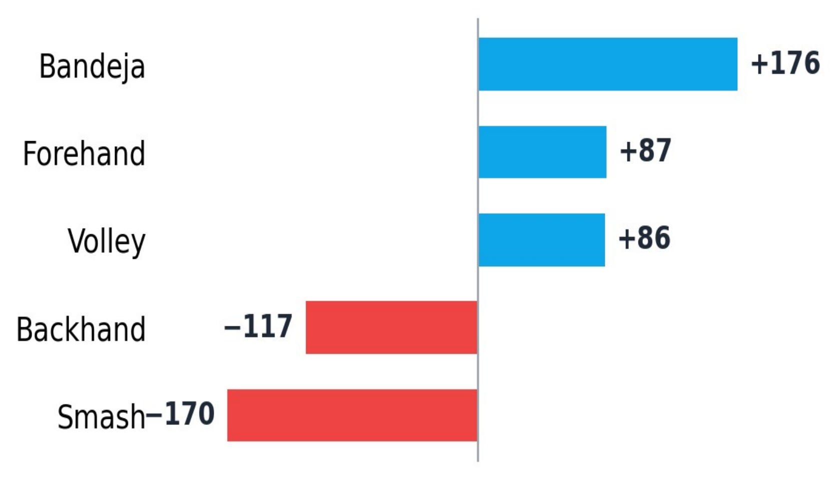
Partner win-rate bars



Win rate per partner; bar length is the fastest channel for comparison (Luca 80% best, Davide 20% worst).

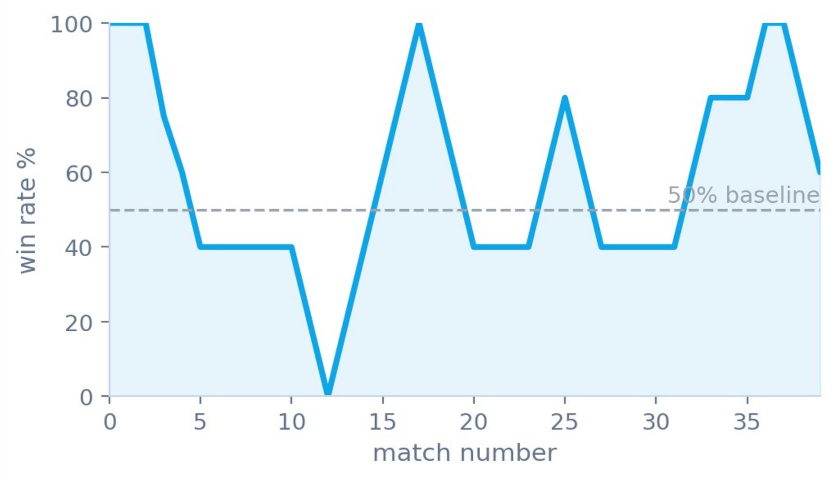
The charts in the app (2/4)

Shot DNA (diverging bars)



Winners minus errors per shot around a zero line - strengths point right (Bandeja +176), weaknesses left (Smash -170).

Rolling win-rate line

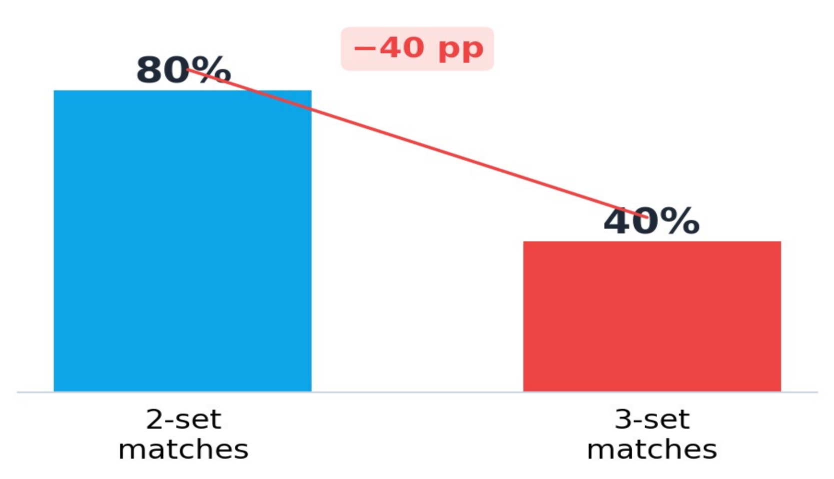


A 5-match rolling mean against a 50% baseline smooths single-match noise to reveal the real trend.

4.7 CHART GALLERY

The charts in the app (3/4)

Set-by-set fatigue



Two-set vs three-set win rate, with the -40-point fatigue gap annotated directly on the chart.

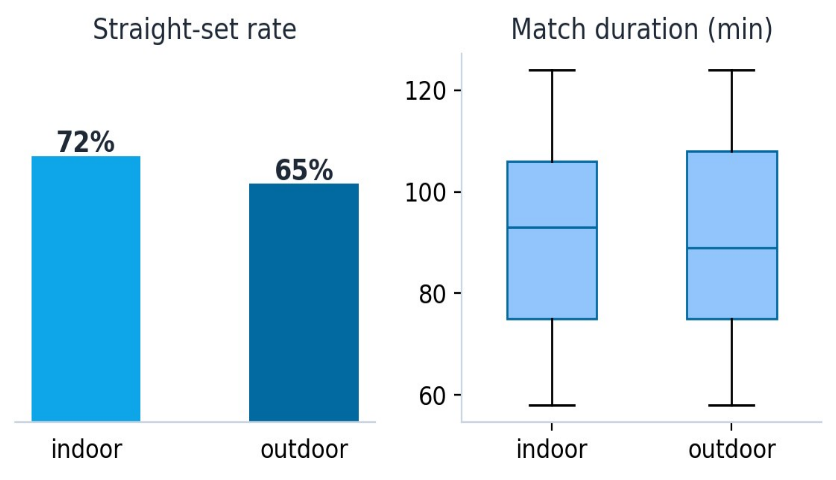
Pro rankings + form sparklines

#	Player	Points	Form
1	Arturo Coello	11,200	
2	Agustin Tapia	11,050	
3	Ale Galan	9,300	
4	Federico Chingotto	9,180	
5	Juan Lebron	7,600	
6	Franco Stupaczuk	7,480	
7	Martin Di Nenno	6,750	
8	Paquito Navarro	6,400	
9	Mike Yanguas	6,120	
10	Momo Gonzalez	5,980	
11	Jon Sanz	5,740	
12	Lucas Bergamini	5,300	

A ranking is inherently tabular; a tiny form sparkline per row adds momentum without a spaghetti plot.

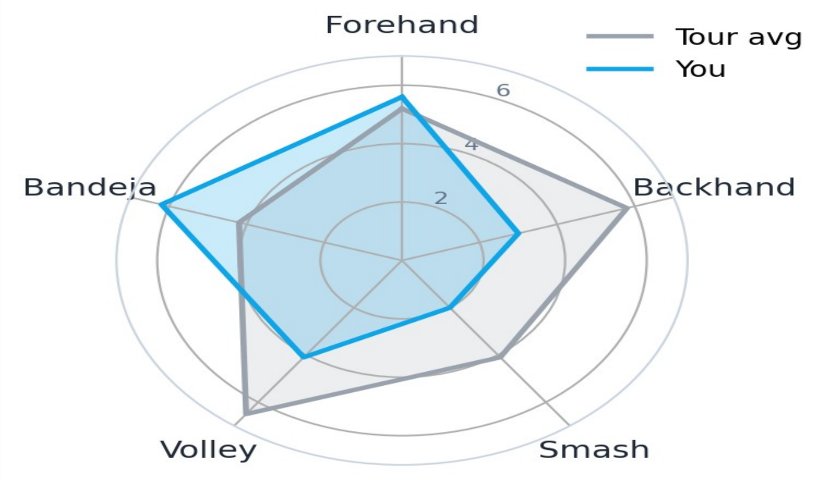
The charts in the app (4/4)

Surface effect (small multiples)



Indoor vs outdoor panels show overlapping distributions - surface is a weak predictor, so it is a filter, not a headline.

Shot-profile radar



Your average winners per shot overlaid on the tour reference across five shot types - the shape gap is the story.

PART 5

Technical Implementation

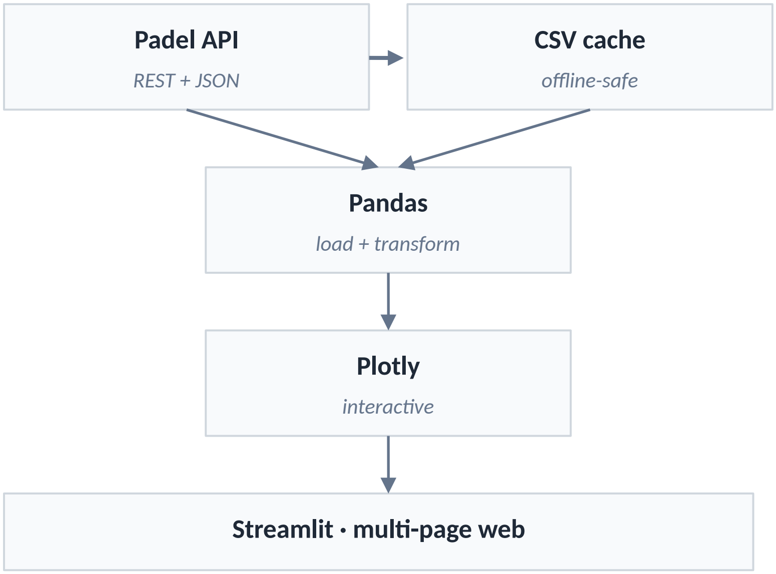
Python · Streamlit · Plotly

What's under the hood

STACK

`Python 3.11`	Language
`Streamlit 1.32`	Multi-page web app framework
`Pandas 2.x`	Data wrangling, groupby, rolling means
`Plotly 5.x`	Interactive charts (hover, zoom, brush)
`Requests`	Padel API client

ARCHITECTURE

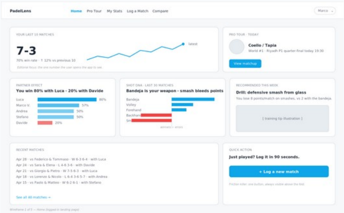


PART 6

Demo & Conclusions

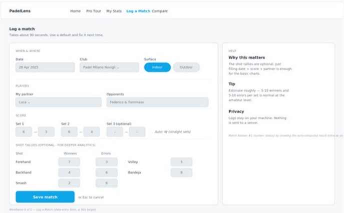
Live walkthrough · Future work

Five-step walkthrough I'll show you next



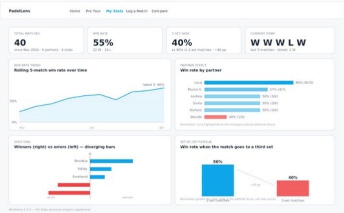
1 · Home

Open the app. Last 10: 7-3. Partner card shows Luca on top, Davide red.



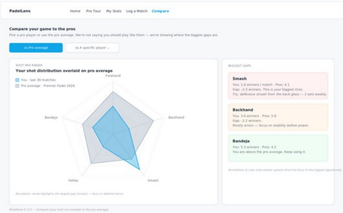
2 · Log Match

Add today's match in 30 seconds. Watch the auto-result update.



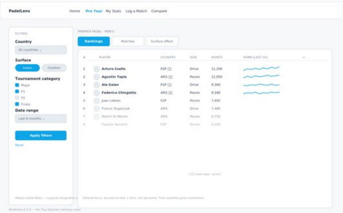
3 · My Stats

Filter to last 60 days. Read the partner & fatigue panels.



4 · Compare

Toggle to a specific pro. See the radar gap on the smash.



5 · Pro Tour

Filter to indoor majors. Read the form sparkline column.

→ Switch to live app now: streamlit run app.py

What v2 would add

+ **Per-player shot averages**

  Padel API doesn't currently expose them. v2 would scrape them from match-level video annotations.

+ **Sensor integration**

  Modern padel paddles (Babolat Pop, Zepp) push event streams. Plugging them in unlocks the spatial layer deliberately left out of v1.

+ **Multi-user support**

  Currently single-user. Add Streamlit auth + per-user CSV so a club coach can see every student.

+ **Mobile-first redesign**

  Logging is a phone task. Streamlit's mobile layout is OK; a native wrap would be better.

+ **LLM-generated weekly digest**

  "This week your bandeja was on; your smash bled 12 points. Drill X."

THANK YOU

Questions?

App code, wireframes, data and these slides bundled in the Padel-Project repository - available on request.

"You don't have to feel your game - you can see it."